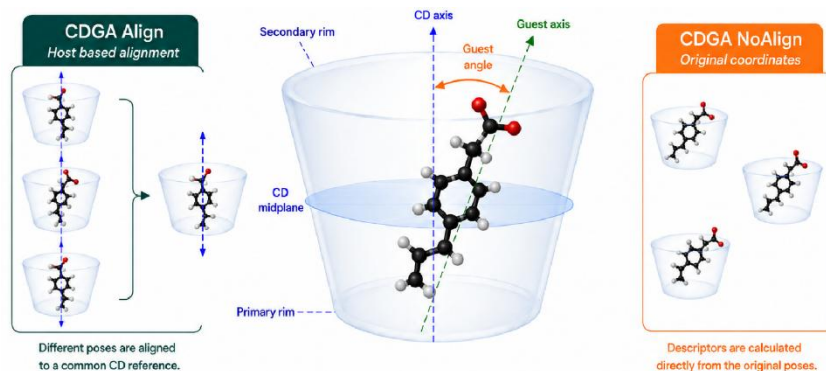


CDGA TOOL

Cyclodextrin Guest Analysis in KNIME



User Manual

OPTIONAL HOST CHARACTERIZATION

Pyranose coordinates
XYZ files for Cremer-Pople
ring puckering analysis

Glycosidic torsion angles
 ϕ and ψ

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1. Introduction

The purpose of this user manual is to provide practical instructions for installing, running, and interpreting CDGA workflows. The associated publication provides scientific reasoning, a comprehensive description of the methodology, validation processes, and comparative analyses. To keep the manual self-contained, essential concepts such as guest depth, guest angle, guest orientation, guest RMSD, and atom-matching levels are briefly summarized.

The installation procedure is included in the **README.md** file in the GitHub repository, however, this document includes pictures to facilitate workflow execution for users without prior experience with the KNIME analytical platform. To provide additional support, the configuration and workflow execution procedures were recorded, and short demo videos are available in the GitHub repository.

Although the CDGA Tool, which contains separate KNIME workflows, is distributed under the GPL-3.0 license, to execute the workflows, the installation of third-party extensions and software are required. Third-party software, libraries, and data remain subject to their respective license terms and are not covered by the CDGA license unless explicitly stated otherwise. This includes KNIME Analytical Platform, open-source libraries accessed through their public APIs, including [RDKit](#), NumPy, pandas and [Open Babel](#).

The Python script included in CDGA was created with the assistance of ChatGPT 5.5; the host characterization appended module and PDB splitter code were created with assistance of ChatGPT 5.6 Sol, OpenAI. Code was subsequently reviewed, verified, and validated by the authors.

Users should review the applicable license terms before installing, executing, modifying, or redistributing the CDGA Tool and its associated data.

This user manual is licensed under the Creative Commons Attribution NonCommercial 4.0 International License, CC BY NC 4.0. You may share and adapt this manual for noncommercial purposes, provided that appropriate credit is given to the authors and any changes are indicated.

If you use the CDGA Tool in a published work, please cite the original paper titled: “CDGA: A Host-Based Alignment and Descriptor Analysis Tool for Cyclodextrin Inclusion Complexes” associated with this User Manual.

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Year: 2026

2. Installation and environment configuration

The CDGA Tool has been created in the open-source KNIME Analytical Platform, which can be downloaded and installed from the [official website](https://www.knime.com/downloads) (https://www.knime.com/downloads). The workflows were prepared using Long-Term Support (LTS) version 5.12.0 for Windows; however, they should work on Windows, Linux, and macOS, if the same Python environment and libraries are available. The scripts use:

1. Python standard libraries:
 - i) os
 - ii) re
 - iii) tempfile
 - iv) itertools
2. Scientific libraries:
 - i) numpy
 - ii) pandas
3. Chemistry Toolkit RDKit
4. KNIME Python integration: knime.scripting.io.

To execute the CDGA Tool, the following parts must be installed:

- **Miniconda:**

1. Download and install Miniconda from the [official website](https://www.anaconda.com/download) (https://www.anaconda.com/download):
2. Set up the Python environment:
 - 2.1. Download the **environment.yml** file from the GitHub repository associated with the CDGA Tool
 - 2.2. Open Anaconda Prompt
 - 2.3. Go to the directory containing the **environment.yml** file using the 'cd' command
 - 2.4. Set up the Python environment by running the command:

```
conda env create -f environment.yml
```

- 2.5. After installing all the dependencies, activate the environment using the command:

```
conda activate knime-rdkit
```

```
Downloading and Extracting Packages:
Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#     $ conda activate knime-rdkit
#
# To deactivate an active environment, use
#
#     $ conda deactivate
```

To test whether all packages were installed correctly, open the Anaconda Prompt and run the following commands:

```
conda env list
```

The environment **knime-rdkit** should be listed.

Then, test whether the required libraries are available:

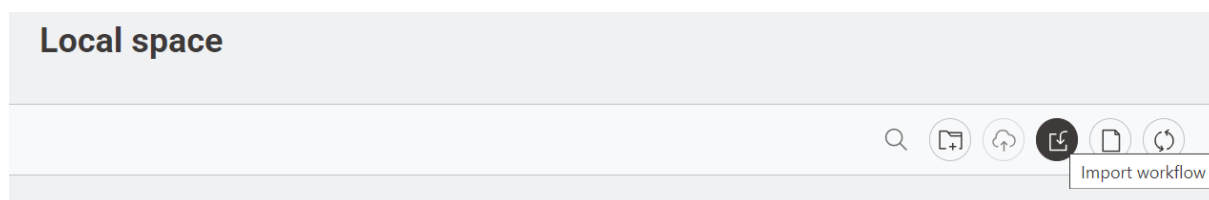
```
python -c "import rdkit, pandas, numpy; print('OK')"
```

If “OK” is printed, the required libraries are installed correctly.

- **KNIME Analytics Platform:**

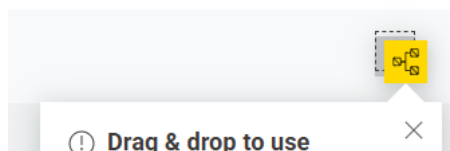
The CDGA Tool has been uploaded to the GitHub repository or can be accessed via the **KNIME Community Hub**.

1. Download the file with **.knwf** extension from the GitHub repository
2. Open KNIME Analytical platform → Local space → Import workflow → find the downloaded file with **.knwf** extension → Open CDGA Tool

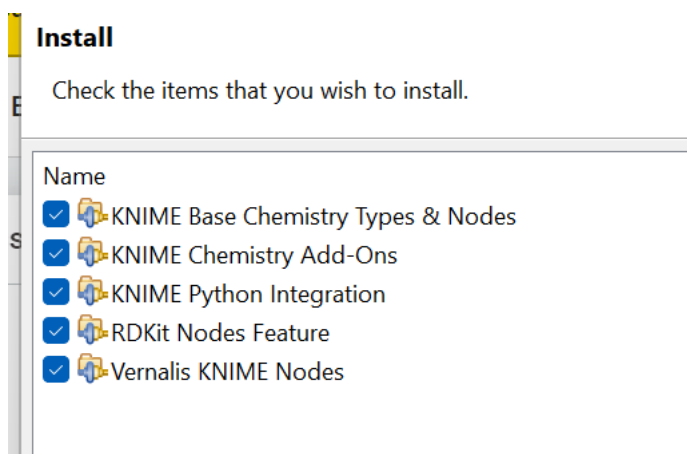
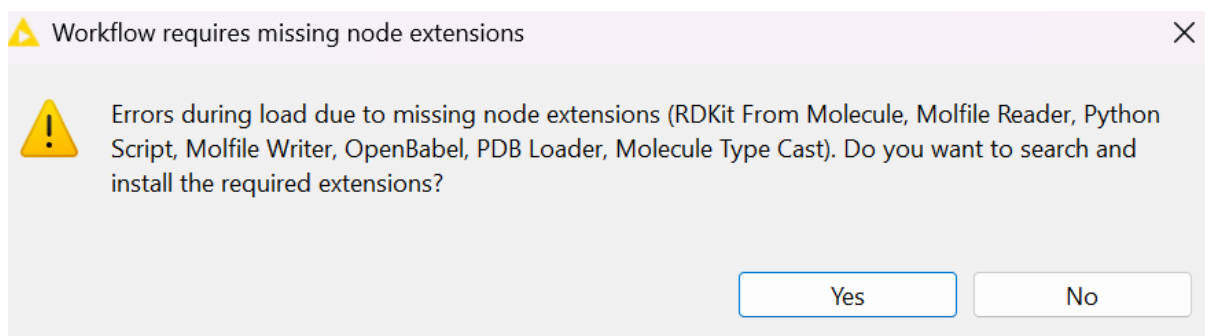


Or access the workflow from the **KNIME Community Hub**:

3. Open KNIME → Open **KNIME Community Hub website** → Use “Drag&drop button or download workflow → find the downloaded .knwf file → Open the CDGA Tool

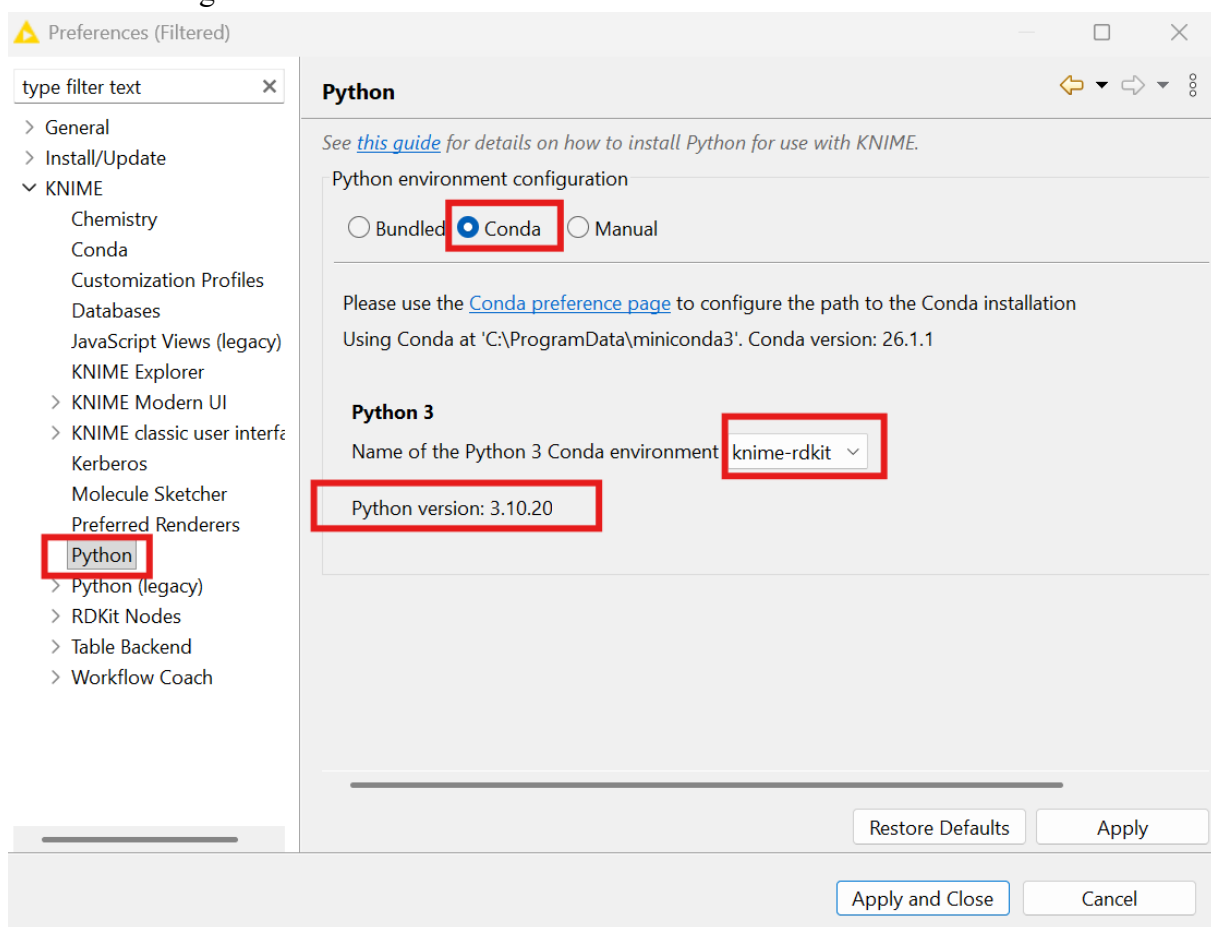


4. The error message should be displayed that the CDGA Tool requires installation of the following extension:



5. Install the required extension.
6. Configure the Python environment in KNIME:
- 6.1. Open the KNIME Analytical Platform and select Preferences → Python:

6.2. Select “**Conda**” in Python environment configuration, and if Conda version is displayed → select “**Name of the Python 3 Conda environment**” as **knime-rdkit**. If the configuration is successfully applied, the Python version should be displayed, as shown in the figure:



or select manually by selecting the path: to `\envs\knime-rdkit\python.exe`

1) to find the path:

Go to Anaconda Prompt (please note that if it has already been closed, the activation of the **knime-rdkit** environment is needed, as it returns to the **base environment** after starting Anaconda Prompt)

a. Test which environment is currently active using the command:

```
conda env list
```

The active environment is marked with *

```
# conda environments:
#
# * -> active
# + -> frozen
base          *
knime-rdkit
```

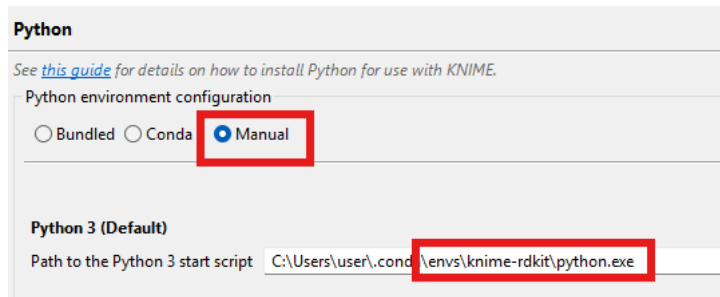
If **base** is active, activate the **knime-rdkit environment** by command:

```
conda activate knime-rdkit
```

After the knime-rdkit activation, check the path to the Python executable by command:

```
where python
```

The path containing “`envs\knime-rdkit\python.exe`” should be provided in KNIME:



Then the Python version should be displayed.

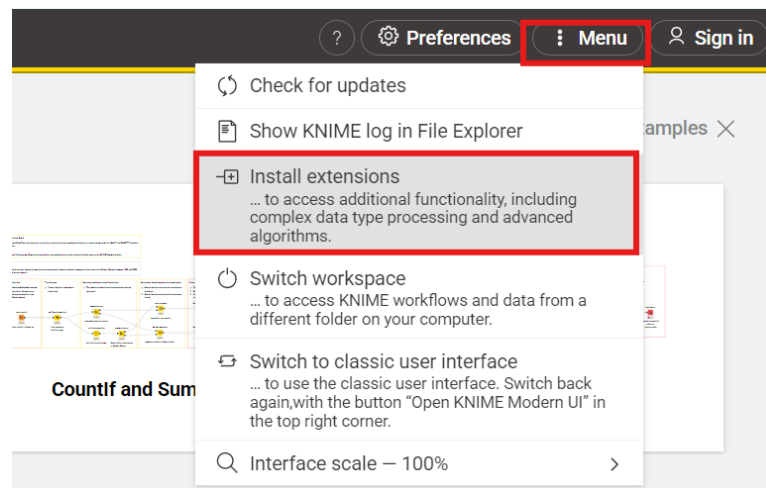
6.3. Click **Apply and Close**.

6.4. After changing the environment, restart KNIME and check whether the preferences are saved.

In case of any error, see the guide provided by KNIME on how to set up the Python environment.

An alternative method of installing extensions:

6.5. Open Knime → Menu → Install extensions

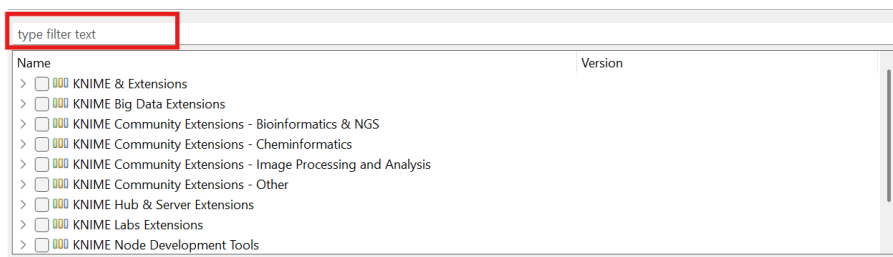


6.6. Search and install the following extensions:

- KNIME Base Chemistry Types & Nodes
- KNIME Python Integration
- RDKit Nodes Feature

Available Software

Check the items that you wish to install.



Optional extensions required for PDB converter only:

- KNIME Chemistry Add-Ons
- Vernalis KNIME Nodes

6.7. Restart KNIME Analytics Platform

Configuration complete. This step only needs to be performed once.

3. Workflow execution in KNIME

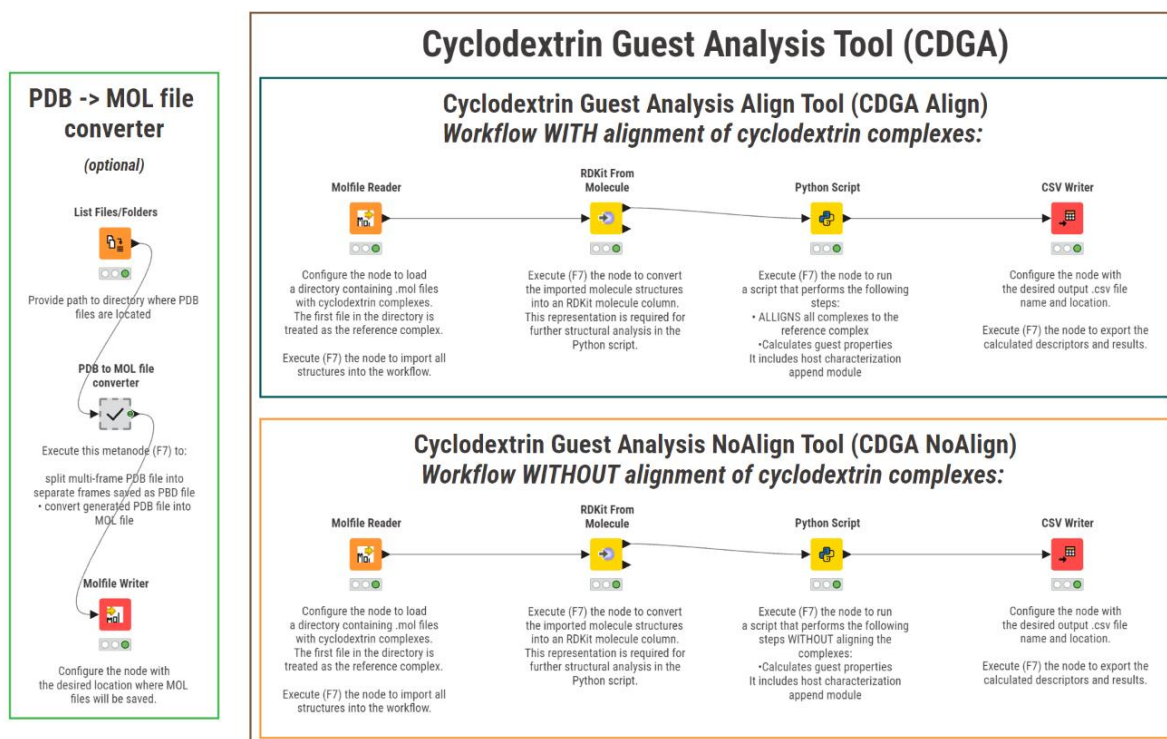
Import the downloaded workflow:

1. Open KNIME → go to Local space → click Import workflow (if not done previously) → find the downloaded file with .knwf extension → Open CDGA Tool
2. Ignore the warning about configuration, if displayed:

CDGA_CD_Guest_Analysis_Tool_host_module 3
 PDB to MOL file converter 3:24
 Python Script 3:24:16
 State has changed from IDLE to CONFIGURED

The CDGA Tool consists of separate workflows: CDGA Align and CDGA NoAlign. Each workflow contains 4 nodes. For user convenience, instructions for operating each node are provided in the descriptions displayed below the corresponding nodes.

Workflow execution should proceed from left to right. Each node should be executed only after the preceding node displays  status. If a node displays  or  light, its configuration and error messages should be checked to identify the source of the problem.

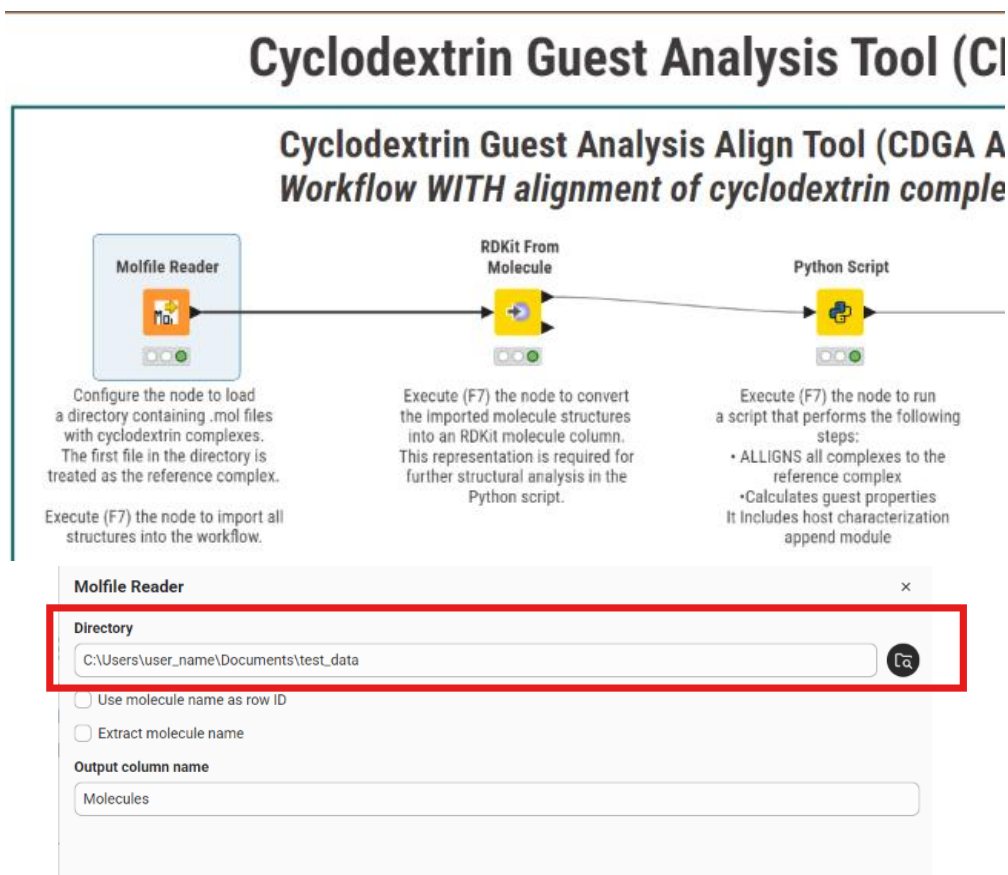


Please note that before configuration, some nodes may display a status other than green because user input is required. In particular, the first node requires the path to the directory containing

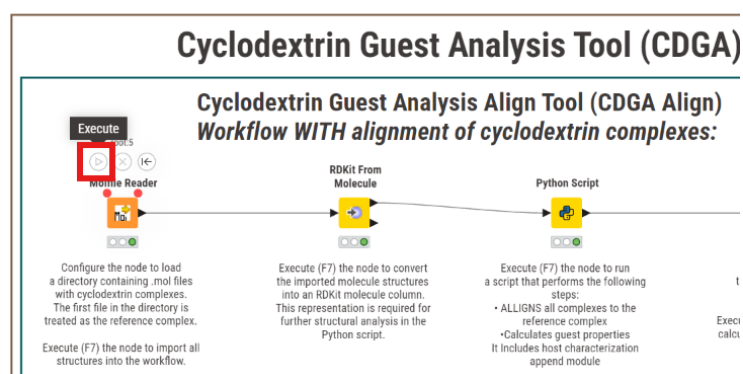
the input MOL files, whereas the final node requires the output directory and name of the resulting CSV file.

1. To configure the first node, right-click the node → Browse the directory containing the MOL files → select the directory → click on **Apply**.

The first MOL file in alphabetical order is treated as the reference complex.



2. Execute the first node:



3. If the node displays a green status indicator, proceed to the next steps and execute the second node (RDKit From Molecule) and the third (Python Script).

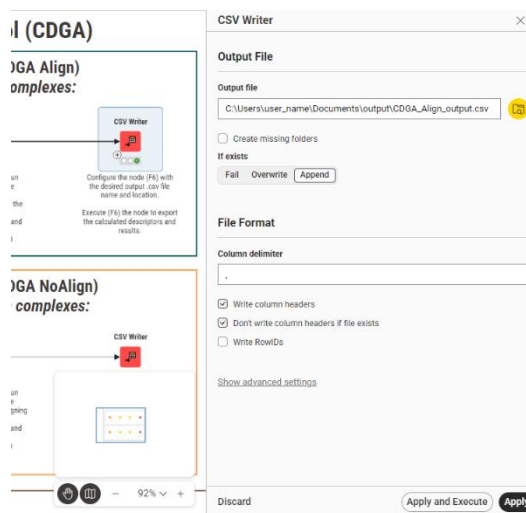
Processed MOL files and CSV files containing atom-matching tables for each structure are generated in a **temporary directory**. The corresponding file paths are listed in the output table.

Users who wish to retain these files should copy them from the temporary directory to a permanent location on their computer.

4. Configure the last node by:

providing the directory and CSV filename for the output table and delimiter. At this stage, the user can decide if the next generated data should be overwritten or appended to the existing file.

Append is set by default.



Then **Execute**.

3.1. Input requirements

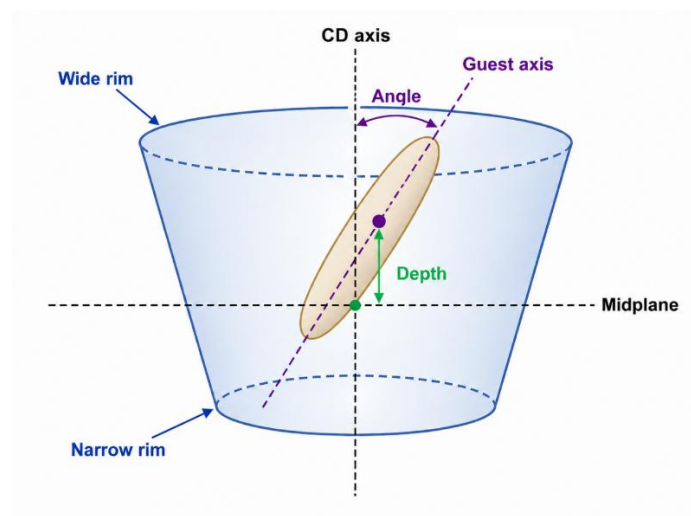
Input structures must be provided in the MOL file format and contain valid three-dimensional coordinates. The CD host is the largest disconnected molecular fragment. The largest remaining fragment containing at least 3 atoms is assigned as the guest.

The input files may also contain additional fragments such as solvent molecules or cations. Therefore, users should verify that the CD is the largest fragment and that the intended guest is the largest remaining fragment satisfying the minimum size criterion. If no suitable guest is detected, the overall_status column reports no_guest. The reference file must contain a guest molecule, however, for tested entries, the presence of a guest is not required, therefore CDGA can be applied for CD analysis.

The host must contain recognizable CD rims to permit the calculation of the CD axis and cavity midpoint. For host refinement in CDGA Align, at least 12 atoms of the cyclic carbon-oxygen framework must be matched between the reference and tested hosts. CDGA was developed for native CDs and validated using α , β , and γ -CD host-guest complexes.

To enable the use of PDB files, including files containing multiple frames, an optional PDB converter workflow was developed. The converter first separates multiple-frame PDB files into individual structures and subsequently converts each structure into MOL format. The converter operates independently of the main CDGA workflow, allowing users to inspect and verify the generated MOL files before further analysis.

4. Methods and calculated properties



Schematic representation of the definitions of the guest depth and guest angle. The purple and green dots represent the geometric centres of the guest heavy atoms and cavity midpoints, respectively. The graphical scheme was generated with the assistance of AI and subsequently reviewed by the authors for its scientific accuracy.

4.1. CD and Guest axis

The largest fragment in the MOL file is treated as the host. Based on the SMILES Arbitrary Target Specification (SMARTS), wide and narrow rims are identified.

SM_O6 = "[CH2][OX2]" → narrow rim (with -CH₂OH group)

SM_SEC = "[CH][OX2]" → wide rim (with -CHOH group)

The CD axis is defined as the vector connecting the geometric centers of the carbon and oxygen atoms forming the narrow primary rim and the wide secondary rim of the host molecule and constitutes the z-axis of the CD-based coordinate frame. Hydrogen atoms are not included in assessing rim centroids. The x-axis is obtained by principal component analysis (PCA) of the primary rim atoms (without hydrogen atoms) projected onto the plane perpendicular to the z-axis. The y-axis is perpendicular to the z and x-axes. In CDGA Align, the reference structure defines the common coordinate frame, and each tested complex is transformed so that its host centroid and local coordinate axes correspond to those of the reference. The host centroid is defined as the geometric center of all atoms of the largest fragment in the MOL file (including hydrogen atoms). The guest axis is defined as the first principal axis obtained by PCA of the heavy-atom coordinates of the guest molecule. The guest axis used for the angle calculation should not be confused with the farthest matched atom pair used for the guest orientation assignment.

4.2. Host alignment

In CDGA Align, both the standard CD frame and an alternative frame rotated by 180° are evaluated to determine whether the entire complex is inverted relative to the reference. This procedure ensures that the narrow and wide rims of the tested complex are oriented consistently with those of the reference complex. After alignment, the host orientation is checked again to confirm correct rim placement. The guest molecule does not influence the host-alignment procedure.

After the correct CD frame has been identified, the script searches for a maximum common substructure (MCS) between the CD cores of the reference and tested complexes. The CD core is defined as the ring carbon and oxygen atoms of the host, and all non-ring atoms, such as hydroxyl groups, are excluded.

The script initially performs a strict MCS search based on matching elements and bond orders, with ring atoms restricted to matching the other ring atoms. If the resulting MCS contains fewer than 12 atoms, a relaxed search is performed. In the relaxed search, atom elements must still match, whereas bond types and ring membership restrictions are ignored. If no sufficiently large common core is identified, the workflow reports a host refinement failure status.

All combinations of MCS atom mappings between the reference and tested CD cores are evaluated using Kabsch fitting. The mapping producing the lowest fitted host RMSD is selected, and the corresponding rigid-body transformation is applied to the entire tested complex. A numerical tolerance of 10^{-6} is used when comparing alternative host-atom mapping with nearly identical RMSD values. During host refinement, Kabsch fitting superimposes the matched CD core atoms onto the reference structure. As part of this transformation, the centroid of the matched host core atoms in the tested structure is placed at the centroid of the corresponding reference atoms.

4.3. Guest depth

Guest depth is defined as the signed displacement of the geometric center of the guest heavy atoms from the CD cavity midpoint, measured along the CD axis toward the narrow rim. The midpoint is defined as the halfway point between the primary and secondary rim centroids. Positive values indicate displacement toward the narrow rim, negative values indicate displacement toward the wide rim, and values near zero indicate that the guest center is near the CD midplane. Guest depth is defined as the projection of the guest heavy-atom centroid onto the CD cavity axis relative to both the tested CD midpoint and the reference CD midpoint, enabling direct comparison of guest positions between complexes.

4.4. Guest Angle

The guest angle is defined as the angle between the local CD and the principal molecular axis of the guest, with values ranging from 0° to 90°. Values close to 0° indicate that the guest axis is approximately parallel to the CD axis, whereas values close to 90° indicate an approximately perpendicular guest to CD axis arrangement.

4.5. Atom mapping and RMSD calculation

Atom mapping of guest heavy atoms is performed using 4 matching levels: **full_guest_strict**, **full_guest_resonance_relaxed**, **full_guest_topology_relaxed** and **mcs**. If the criteria for a given level are not satisfied, the next, less restrictive level is applied:

Matching level	Criteria	Differences allowed
full_guest_strict	Full guest match. The atom type, atom aromaticity, ring membership, formal charge, hydrogen count, number and type of heavy atom neighbours, bond order, and bond aromaticity must be consistent.	No major differences in chemical representation are allowed. This is the most restrictive category of all.
full_guest_resonance_relaxed	Full guest match. The atom type, atom aromaticity, ring membership, hydrogen count, number of heavy atom neighbours, and neighbouring heavy atom types must be consistent.	Bond order and formal charge differences are allowed. This category is intended for resonance or differences in charge.
full_guest_topology_relaxed	Full guest match after topology simplification. The atom type, ring membership, number of heavy atom neighbors, and neighboring heavy atom types must be consistent.	The bond order, bond aromaticity, atom aromaticity, formal charge, and hydrogen count are not used as matching criteria. All types of bonds are simplified to a single bond.
mcs	No valid full guest match is found. The script searches for the maximum common substructure (MCS) and uses the matched common fragments.	The guests may be chemically different. Only the common substructure is used for the RMSD and orientation analyses.

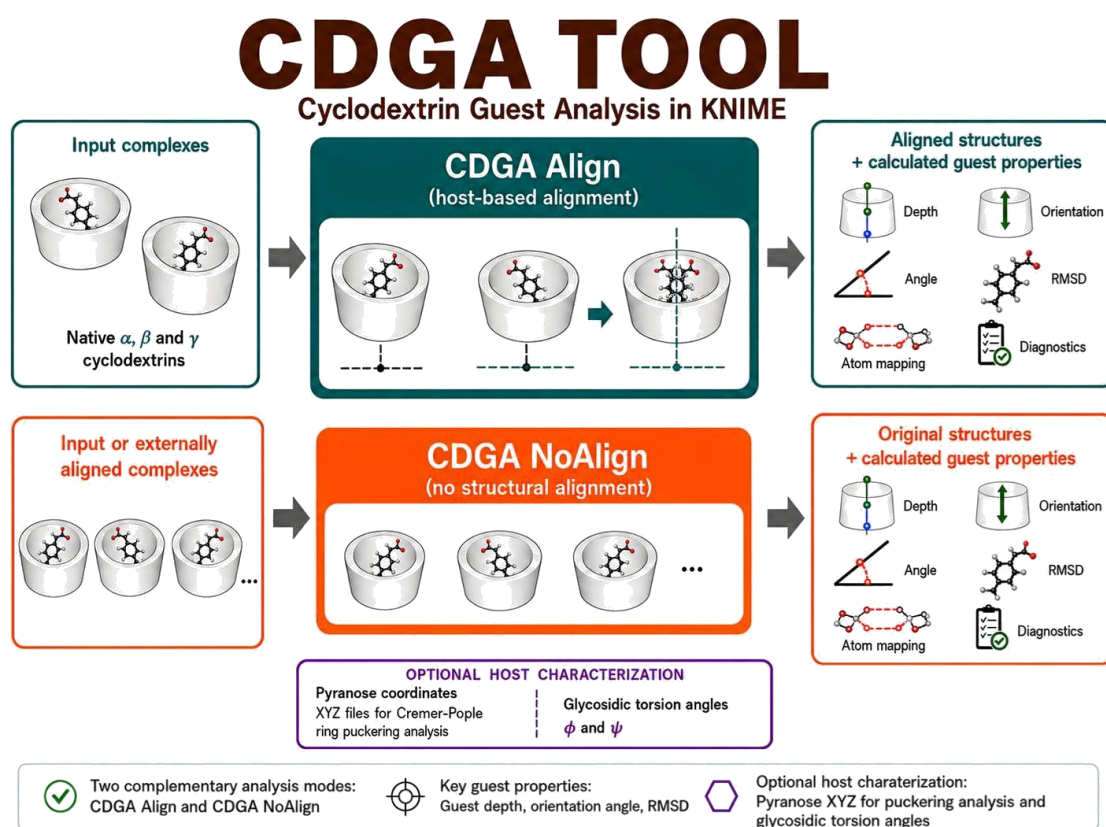
The hydrogen atoms in the external functional groups can be explicitly represented in the input MOL files. When the hydrogen counts and formal charges are consistently assigned in the reference and tested structures, a more stringent matching level may be achieved. However, the hydrogen counts and formal charges should be set correctly, as incorrect settings may affect the matching of guest molecules.

Guest RMSD is calculated using the matched heavy-atom pairs. A numerical tolerance of 10^{-3} Å is used to identify alternative atom mappings with tied RMSD values. The second-lowest RMSD is also reported in the output table, allowing users to assess the uniqueness and reliability of the selected atom mapping.

4.6. Guest orientation

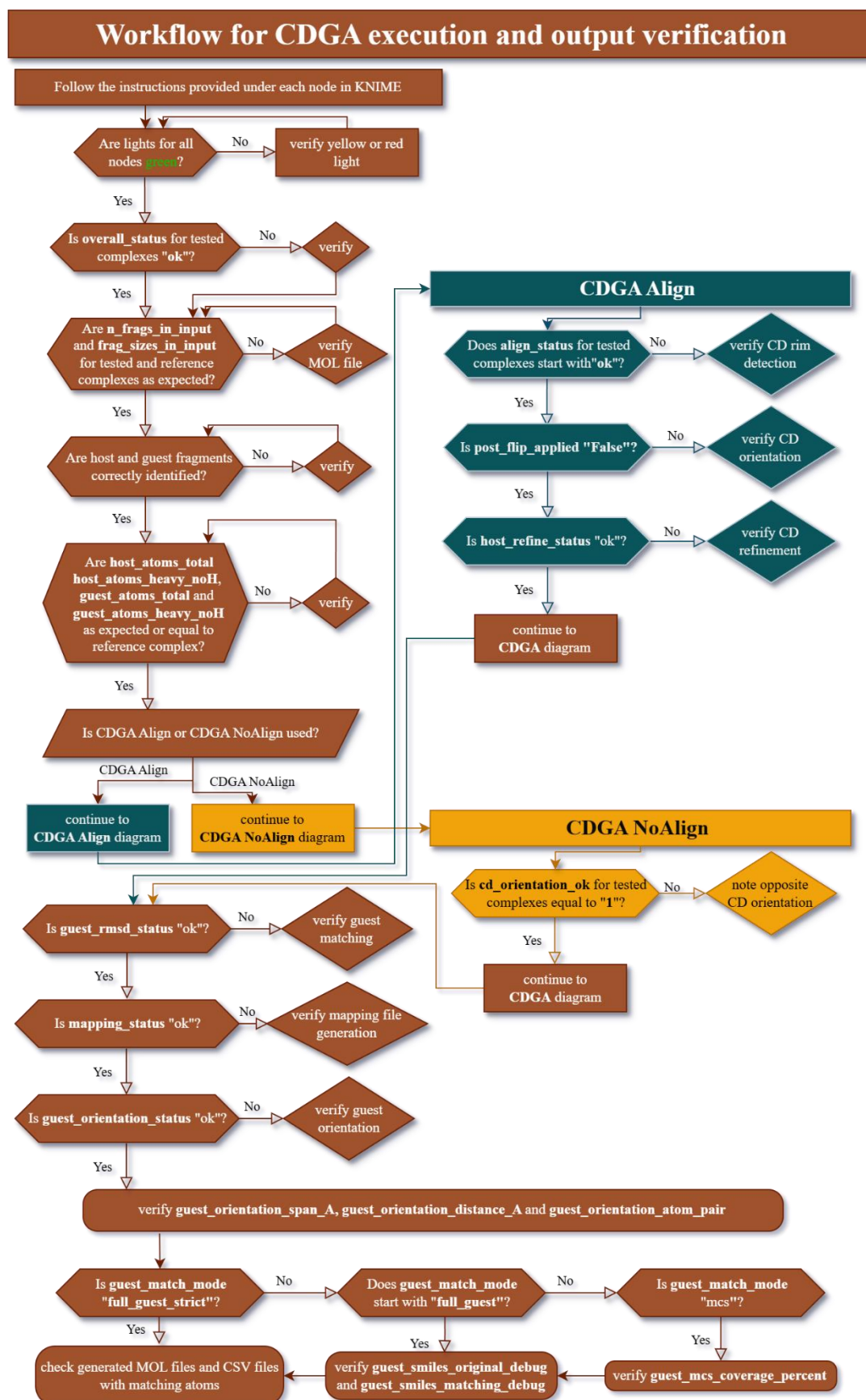
Guest orientation is determined by using a pair of matched heavy guest atoms separated by the greatest distance. The atom pair is selected from the full guest when **full-guest matching** is achieved or from the MCS when the **mcs** matching level is applied. Among the matched guest atoms, the pair with the greatest interatomic distance in the reference structure is selected to define the guest orientation. The two reference atoms are projected onto the reference CD axis to determine which atom lies farther in the positive axis direction. The corresponding atom pair in the tested complex is then projected onto the same reference CD axis. The relative positions of these atoms along the CD axis are compared between the reference and tested complexes to determine whether the guest orientation is preserved or reversed. The corresponding atom pairs for the reference and test complexes are listed in the output table for verification. Each matched atom in the tested complex must have the same element type as its counterpart in the reference complex.

4.7. Summary of CDGA Tool



The graphical summary of CDGA Tool was generated with the assistance of AI and subsequently reviewed by the authors for its accuracy.

5. Workflow diagnostics



6. CDGA Align – output table

Column name	Description	Meaning	Possible options or examples
entry_name	Original row identifier from the input table	Used to keep the link between the input structure and the calculated results	row ID from previous nodes, usually MOL file name
saved_mol_path	Path where aligned complexes exported as .mol files are saved	Used to locate aligned complex files for further visualization	by default, the path to the system temporary directory: Windows: C:\Users\USER_NAME\AppData\Local\Temp\CDGA_Tool_processed_molfiles\CDGA_Align_molfiles\mol_file_name.mol Linux: /tmp/CDGA_Tool_processed_molfiles/CDGA_Align_molfiles/mol_file_name.mol
mapping_csv_path	Path to CSV file with guest atom mapping	Allows inspection of guest atom correspondence	path in CDGA_Align_molfiles/Matching_atoms; reference row: missing value
mapping_status	Status of mapping file creation	Indicates if mapping was generated successfully	ref - reference row ok - mapping created no_guest - no guest found no_mcs - no common substructure mcs_match_failed - mapping failed
overall_status	Overall result status	Summary of full workflow success	ref - reference row ok - all steps successfully performed align_failed - issue at the step with the alignment of the complexes host_refine_* - issue at the step with host refinement guest_match_* - issue at the step with guest matching mapping_* - issue at the step with mapping

Column name	Description	Meaning	Possible options or examples
align_status	Status of complex alignment	Shows whether alignment to the reference CD frame was successful and if flip correction was applied	ref_noflip - reference structure, no flip applied ref_postflip - theoretical flipped reference ok_axis_frame_noflip - successful alignment using standard-frame, no post-alignment flip needed ok_axis_frame_postflip -successful alignment using standard-frame and post-alignment flip applied ok_axis_frame_180_noflip - successful alignment using the 180° alternative frame, no post-alignment flip needed ok_axis_frame_180_postflip - successful alignment, using the 180° alternative frame and post alignment flip applied rims_not_found_noflip - alignment not possible because required rim atoms were not detected rims_not_found_postflip - mostly theoretical option, alignment failed, but flip still applied afterwards
post_flip_applied	Indicates if a 180° flip correction was applied	Ensures consistent CD axis direction. After alignment, the script compares the CD axis direction to the reference. If opposite, a 180° flip in the reference frame is applied	True or False: If post_flip_applied is True, the suffix "_postflip" is added to align_status. If it is False, the suffix "_noflip" is added.
host_refine_status	Status of host refinement	Indicates if cyclodextrin core refinement succeeded	ref - reference row ok - refinement successful no_host_core - core not detected host_core_mcs_too_small - insufficient overlap host_core_match_failed - mapping failed kabsch_failed - transformation failed
host_atoms_used_in_refine	Number of atoms used in refinement	Indicates how many heavy atoms from the cyclodextrin core structure were used to refine the alignment. Higher values mean more reliable host alignment	integer or for reference row: 0

CDGA Align – output table

Column name	Description	Meaning	Possible options or examples
host_rmsd_before_refine	RMSD before refinement	Measures how well the cyclodextrin core (skeleton) of the tested complex matches the reference before applying refinement. Indicates initial host misalignment	numeric value in Å or for reference row: missing value
host_rmsd_after_refine	RMSD after refinement	Measures how well the cyclodextrin core (skeleton) matches the reference after refinement. Indicates final host alignment quality	numeric value in Å or for reference row: missing value
n_frgs_in_input	Number of disconnected fragments in the input structure	Checks whether the host and guest are separated and how many independent fragments are present in the structure	For a typical cyclodextrin host-guest complex, it should be 2; values above 2 may indicate solvent, ions, or a fragmented guest
frag_sizes_in_input	Sizes of all detected fragments	Comma-separated number of atoms in each fragment	example: 28, 147, 3 means three fragments with 28, 147, and 3 atoms
host_atoms_total	Total number of atoms in the host fragment	Number of atoms in the largest fragment. The script identifies the largest fragment as the host.	integer, for native β-cyclodextrin: 147
host_atoms_heavy_noH	Number of heavy atoms in the host fragment	Host size excluding hydrogens	integer, for native β-cyclodextrin: 77
host_core_atoms_heavy_noH	Number of heavy atoms in CD core	Indicates how many non-hydrogen atoms from the cyclodextrin ring skeleton were identified and used as the core for the host refinement. The core includes the ring atoms and glycosidic bridge oxygens.	integer value, for native β-cyclodextrin: around 49; lower values may indicate incomplete core detection or structural problems
guest_atoms_total	Total number of atoms in the guest fragment	Number of atoms in the second-largest fragment. The script selects the largest non-host fragment with at least three atoms as the guest.	integer; 0 means no valid guest fragment was detected, e.g. only one fragment present or all additional fragments are smaller than 3 atoms
guest_atoms_heavy_noH	Number of heavy atoms in the guest molecule	Number of atoms in the second-largest fragment, excluding hydrogen atoms.	integer; 0 means no valid guest fragment was detected
guest_depth_A_from_midpoint_to_O6	Guest depth measured along the local CD axis.	Projection of the guest center onto the axis from the midpoint between the rims toward the narrow (O6) rim using the current tested complex geometry.	numeric value in Å, positive value -guest shifted toward narrow ring negative value - guest shifted toward the opposite side. Missing value if calculation could not be performed

Column name	Description	Meaning	Possible options or examples
guest_depth_A_from_ref_midpoint_to_O6	Guest depth measured relative to the reference CD midpoint and direction.	Projection of the guest center onto the reference axis from the reference midpoint toward the reference narrow (O6) rim. Enables comparison across structures	numeric value in Å, positive value -guest shifted toward narrow ring negative value - guest shifted toward the opposite side. Missing value if calculation could not be performed
guest_angle_deg_vs_axis	Angle between guest axis and CD axis	The guest principal axis is estimated using PCA of the heavy atom coordinates and compared to the local cyclodextrin axis. Describes how the guest is oriented relative to the cavity axis	value in degrees (°), range from 0 to 90; missing value if guest axis cannot be determined; examples: 0 = guest axis parallel to CD axis, 90 = guest axis perpendicular to CD axis
guest_rmsd_noH	RMSD of the tested guest compared with the reference guest using heavy atoms only	Measures how well the tested guest aligns with the reference guest based on the matched heavy atom coordinates after alignment.	numeric value in Å or for reference row: missing value; lower values indicate higher similarity
guest_rmsd_status	Status of RMSD calculation	Indicates whether the guest molecule could be matched to the reference and the RMSD calculated.	ref: reference row ok: RMSD calculated successfully no_guest: no guest fragment identified no_heavy_guest: guest contains no heavy atoms no_mcs: no common substructure found mcs_match_failed: no valid atom mapping obtained
guest_atoms_used_in_rmsd	Number of atoms used for RMSD calculation	Indicates the number of heavy atoms used to compute the RMSD between the tested and reference guests. Full match uses all heavy atoms, MCS uses only the common subset	integer value

CDGA Align – output table

Column name	Description	Meaning	Possible options or examples
guest_match_mode	Matching method used	Defines how strictly the tested guest was matched to the reference guest before RMSD calculation	ref - reference row 4 levels: full_guest_strict - exact match of the whole guest; atoms and their bonding must be the same, representing chemically identical structures full_guest_resonance_relaxed - exact match of the whole guest; allows differences in bond order or resonance (e.g. aromatic vs double bond representation), chemically equivalent structures still match full_guest_topology_relaxed - match of the whole guest based only on atom connectivity; bond types and aromaticity are ignored, useful when structures are inconsistently represented but have the same overall connectivity mcs - partial match based on the largest common substructure; only the shared part of the guest is used; applied when guests are chemically different
guest_full_match_coverage_percent	Coverage of a full guest match	Percentage of tested guest heavy atoms used for full guest matching. Indicates whether the entire guest was successfully matched to the reference	100.0 for full match, missing value otherwise; for reference row: missing value
guest_mcs_coverage_percent	Coverage of the MCS match	Percentage of tested guest heavy atoms used in MCS-based matching when the full guest match fails. Indicates how much of the guest is represented in the partial match	numeric value, e.g., 60.0, 85.7; missing for full match; for reference row: missing value
guest_orientation	Guest orientation relative to reference	Indicates whether guest orientation inside the CD cavity matches the reference based on the selected atom pair along the CD axis	1 same orientation as reference, 0 reversed orientation; for reference row: missing value

Column name	Description	Meaning	Possible options or examples
guest_orientation_status	Status of orientation calculation	Indicates whether guest orientation could be determined	ref - reference row ok - orientation successfully determined no_guest - no guest fragment identified no_mcs - no matching atoms available no_endpoints - insufficient atoms to define orientation
guest_orientation_span_A	Distance between selected reference atoms	Distance between the two farthest matched reference guest atoms used to define orientation	numeric value in Å
guest_orientation_distance_A	Distance in the tested molecule	Distance between the corresponding atom pair in the tested guest	numeric value in Å
guest_orientation_atom_pair	Atom pair used for orientation	Shows which matched atom pairs between the tested and reference guest were used to determine orientation	example: A: C1→C4 B: O8→O10, meaning tested atom C1 corresponds to reference C4 and tested atom O8 corresponds to reference O10; these pairs define guest orientation, for reference row: ref
guest_orientation_atoms_used	Number of atoms used in orientation calculation	Number of atoms contributing to orientation determination, usually equal to the number of atoms used in RMSD	integer value, for reference row: 0
match_ref_matches_count	Number of reference substructure matches	Indicates how many possible matches were found in the reference guest	integer value, for reference row: missing value
match_prb_matches_count	Number of probe matches	Indicates how many possible matches were found in the tested guest	integer value, for reference row: missing value
match_total_pairings	Total number of match combinations tested	Number of all reference to probe mapping combinations evaluated during matching	integer value, for reference row: missing value
match_mapping_is_unique	Indicates whether the best RMSD-based atom mapping is unique	Shows if only one atom mapping gives the lowest RMSD or multiple equivalent mappings exist	True or False; True means only one mapping produced the lowest RMSD; False means multiple mappings give the same lowest RMSD within tolerance (10^{-3} Å); for reference row: missing value
match_best_rmsd_is_tie	Indicates whether multiple atom mappings share the same best RMSD	Identifies ambiguity in atom correspondence due to symmetry or equivalent mappings	True or False; True means multiple mappings have the same lowest RMSD within tolerance (10^{-3} Å); False means a single best mapping; for reference row: missing value

CDGA Align – output table

Column name	Description	Meaning	Possible options or examples
match_best_rmsd_second_best_delta	Difference between the best RMSD and the next distinct (non-tied) RMSD values.	Difference between the best RMSD and the next non-equivalent RMSD value (excluding mappings considered equal within a tolerance of 10^{-3} Å)	numeric value in Å; large value indicates clear best mapping; small value indicates ambiguity; for reference row: missing value
guest_smiles_original_debug	SMILES comparison for original structures	Compares the canonical SMILES of reference (REF) and the tested guest (PRB) before normalization	format: REF=... PRB=...
guest_smiles_matching_debug	SMILES after relaxed normalization	Compares the canonical SMILES of the reference (REF) and the tested guest (PRB) after normalization, used for matching. The level of normalization depends on the guest_match_mode (e.g., bond types and aromaticity may be simplified for relaxed matching)	format: REF=... PRB=...
guest_ref_heavy_noH	Number of heavy atoms in the reference guest	Provides reference guest size used for comparison	integer value

7. CDGA NoAlign – output table

Column name	Description	Meaning	Possible options or examples
entry_name	Original row identifier from the input table	Used to keep the link between the input structure and the calculated results	row ID from previous nodes, usually MOL file name
saved_mol_path	Path where original non-aligned complexes are saved as .mol files	Used to locate exported structures for visualization	by default, the path to the system temporary directory: Windows: C:\Users\USER_NAME\AppData\Local\Temp\CDGA_Tool_processed_molfiles\CDGA_NoAlign_molfiles\mol_file_name.mol Linux: /tmp/CDGA_Tool_processed_molfiles/CDGA_NoAlign_molfiles/mol_file_name.mol
mapping_csv_path	Path to CSV file with guest atom mapping	Allows inspection of guest atom correspondence	path in CDGA_NoAlign_molfiles/Matching_atoms; reference row: missing value
mapping_status	Status of mapping file creation	Indicates if mapping was generated successfully	ref-reference row ok - mapping created no_guest - no guest found no_mcs - no common substructure mcs_match_failed - mapping failed probe_cd_axis_missing - CD axis missing in tested structure
overall_status	Overall result status	Summary of full workflow success	ref - reference row ok - all steps successfully performed guest_match_ - issue with guest matching mapping_ - issue with mapping
n_frags_in_input	Number of disconnected fragments in the input structure	Checks whether the host and guest are separated and how many independent fragments are present in the structure	for a typical cyclodextrin host-guest complex, it should be 2; values above 2 may indicate solvent, ions, or fragmented guest

Column name	Description	Meaning	Possible options or examples
frag_sizes_in_input	Sizes of all detected fragments	Comma-separated number of atoms in each fragment	example: 28, 147, 3 means three fragments with 28, 147 and 3 atoms
host_atoms_total	Total number of atoms in the host fragment	Number of atoms in the largest fragment, selected as cyclodextrin host	integer, for native β -cyclodextrin: 147
host_atoms_heavy_noH	Number of heavy atoms in the host fragment	Host size excluding hydrogens	integer, for native β -cyclodextrin: 77
guest_atoms_total	Total number of atoms in the guest fragment	Number of atoms in the second-largest fragment. The script selects the largest non-host fragment with at least 3 atoms as the guest.	integer value; 0 means no valid guest fragment was detected, e.g., only one fragment present or all additional fragments are smaller than 3 atoms
guest_atoms_heavy_noH	Number of heavy atoms in the guest molecule	Number of atoms in the second largest fragment excluding hydrogens.	integer; 0 means no valid guest fragment was detected
guest_ref_heavy_noH	Number of heavy atoms in the reference guest	Provides reference guest size used for comparison	integer value
cd_axis_dot_to_ref	Dot product between the tested CD axis and the reference CD axis	Shows whether the cyclodextrin axis of the tested structure points in the same or opposite direction as the reference	value from about minus 1 to 1; close to 1 means same direction; close to minus 1 means opposite direction; missing if CD axis cannot be calculated
cd_orientation_ok	Direction agreement of the tested CD axis with the reference CD axis	Indicates if the tested cyclodextrin axis has the same general direction as the reference	1 - same direction or dot product ≥ 0 , 0 - opposite direction, missing - CD axis could not be calculated
guest_depth_A_from_midpoint_to_O6	Guest depth measured along the local CD axis.	Projection of the guest center onto the axis from the midpoint between the rims toward the narrow (O6) rim using the currently tested complex geometry.	numeric value in Å, positive value -guest shifted toward narrow ring; negative value, the guest shifted toward the opposite side. Missing value if calculation could not be performed
guest_depth_A_from_ref_midpoint_to_O6	Guest depth measured relative to the reference CD midpoint and direction	Projection of the guest center onto the reference axis from the reference midpoint toward the reference O6 rim. Since structures are not aligned, interpret carefully	numeric value in Å, positive value -guest shifted toward narrow ring; negative value, the guest shifted toward the

Column name	Description	Meaning	Possible options or examples
			opposite side. Missing value if calculation could not be performed
guest_angle_deg_vs_axis	Angle between guest axis and CD axis	The guest principal axis is estimated using PCA of the heavy atom coordinates and compared to the local CD axis. Describes how the guest is oriented relative to the cavity axis.	value in degrees (°), range from 0 to 90; missing value if guest axis cannot be determined; examples: 0 = guest axis parallel to CD axis, 90 = guest axis perpendicular to CD axis
guest_rmsd_noH	RMSD of the tested guest compared with the reference guest using heavy atoms only	Measures the difference between matched guest heavy atom coordinates in the original non-aligned coordinates	numeric value in Å or for reference row: missing value; lower values indicate higher similarity
guest_rmsd_status	Status of RMSD calculation	Indicates whether the guest molecule could be matched to the reference and the RMSD calculated	ref - reference row ok - RMSD calculated successfully no_guest - no guest fragment identified no_heavy_guest - guest contains no heavy atoms no_mcs - no common substructure found mcs_match_failed - no valid atom mapping obtained probe_cd_axis_missing - CD axis missing in tested structure
guest_atoms_used_in_rmsd	Number of atoms used for RMSD calculation	Indicates the number of heavy atoms used to compute the RMSD between the tested and reference guests. Full match uses all heavy atoms, MCS uses only the common subset	integer value
guest_match_mode	Matching method used	Defines how strictly the tested guest was matched to the reference guest before RMSD calculation	ref - reference row 4 levels: full_guest_strict - exact match of the whole guest; atoms and their bonding must be the same, representing chemically identical structures

CDGA NoAlign – output table

Column name	Description	Meaning	Possible options or examples
			full_guest_resonance_relaxed - exact match of the whole guest; allows differences in bond order or resonance (e.g. aromatic vs double bond representation), chemically equivalent structures still match full_guest_topology_relaxed - match of the whole guest based only on atom connectivity; bond types and aromaticity are ignored, useful when structures are inconsistently represented but have the same overall connectivity mcs - partial match based on the largest common substructure; only the shared part of the guest is used; applied when guests are chemically different none - matching not performed
guest_full_match_coverage_percent	Coverage of a full guest match	Percentage of tested guest heavy atoms used in full guest matching. Indicates whether the entire guest was successfully matched to the reference	100.0 for full match, missing value otherwise; for reference row: missing value
guest_mcs_coverage_percent	Coverage of the MCS match	Percentage of tested guest heavy atoms used in MCS-based matching when a full guest match fails. Indicates how much of the guest is represented in the partial match	numeric value, e.g., 60.0, 85.7; missing for full match; for reference row: missing value
guest_orientation	Guest orientation relative to reference	Indicates whether guest orientation matches reference based on the selected atom pair along each molecule's own CD axis	1 same orientation as reference, 0 reversed orientation; for reference row: missing value
guest_orientation_status	Status of orientation calculation	Indicates whether guest orientation could be determined	ref - reference row ok - orientation determined no_guest - no guest identified no_heavy_guest - guest has no heavy atoms

Column name	Description	Meaning	Possible options or examples
			no_mcs - no matching atoms available mcs_match_failed - no valid atom mapping no_endpoints - insufficient atoms probe_cd_axis_missing - CD axis missing
guest_orientation_span_A	Distance between selected reference atoms	Distance between the two farthest matched reference guest atoms used to define orientation	numeric value in Å
guest_orientation_distance_A	Distance in the tested molecule	Distance between the corresponding atom pair in the tested guest	numeric value in Å
guest_orientation_atom_pair	Atom pair used for orientation	Shows which matched atom pairs between the tested and reference guest were used to determine orientation	example: A: C1→C4 B: O8→O10, meaning tested atom C1 corresponds to reference C4 and tested atom O8 corresponds to reference O10; these pairs define guest orientation, for reference row: ref
guest_orientation_atoms_used	Number of atoms used in orientation calculation	Number of atoms contributing to orientation determination, usually equal to the number of atoms used in RMSD	integer value, for reference row: 0
match_ref_matches_count	Number of reference substructure matches	Indicates how many possible matches were found in the reference guest	integer value, for reference row: missing value
match_prb_matches_count	Number of probe matches	Indicates how many possible matches were found in the tested guest	integer value, for reference row: missing value
match_total_pairings	Total number of match combinations tested	Number of all reference to probe mapping combinations evaluated during matching	integer value, for reference row: missing value
match_mapping_is_unique	Indicates whether the best RMSD-based atom mapping is unique	Shows if only one atom mapping gives the lowest RMSD or multiple equivalent mappings exist	True or False; True means only one mapping produced the lowest RMSD; False means multiple mappings give the same lowest RMSD within tolerance (10^{-3} Å); for reference row: missing value

CDGA NoAlign – output table

Column name	Description	Meaning	Possible options or examples
match_best_rmsd_is_tie	Indicates whether multiple atom mappings share the same best RMSD	Identifies ambiguity in atom correspondence due to symmetry or equivalent mappings	True or False; True means multiple mappings have the same lowest RMSD within tolerance (10^{-3} Å); False means a single best mapping; for reference row: missing value
match_best_rmsd_second_best_delta	Difference between the best RMSD and the next distinct (non-tied) RMSD value.	Difference between the best RMSD and the next non-equivalent RMSD value (excluding mappings considered equal within a tolerance of 10^{-3} Å)	numeric value in Å; large value indicates clear best mapping; small value indicates ambiguity; for reference row: missing value
guest_smiles_original_debug	SMILES comparison for original structures	Compares the canonical SMILES of the reference (REF) and the tested (PRB) guest before normalization	format: REF=... PRB=...
guest_smiles_matching_debug	SMILES after relaxed normalization	Compares the canonical SMILES of the reference (REF) and tested (PRB) guests after normalization for matching. The level of normalization depends on the guest_match_mode (e.g., bond types and aromaticity may be simplified for relaxed matching)	format: REF=... PRB=...

8. Appendix 1: Host Characterization module

Host Characterization appended module has been added to both CDGA Align and CDGA NoAlign to provide additional structural information about the CD host. It is intended for analysis of α , β , and γ -CD containing 6, 7, and 8 pyranose units, respectively. The number of pyranose units is detected automatically from the reference host, and all tested structures are expected to contain the same number of pyranose units as the reference.

The numbering of the identified pyranose units is established from the CD host of the reference complex. Pyranose 1 is assigned to the unit with the lowest original C1 atom index, and the remaining pyranose units are numbered sequentially according to the glycosidic connectivity around the cyclodextrin ring.

For the tested complexes, pyranose numbering is assigned according to the atom correspondence with the reference host, enabling direct comparison of host characterization results between structures.

In CDGA NoAlign, no structural alignment is performed, however, host atom correspondence is still determined to establish consistent pyranose matching with the reference. Therefore, host characterization results from CDGA NoAlign should be interpreted with caution, particularly when comparing structures with substantially different host orientations or conformations.

Host Characterization Appended Module is executed only after completion of the CDGA Align or CDGA NoAlign module, so that any issues encountered during host characterization do not interfere with the main CDGA analysis.

Output files are saved in the corresponding Host_characterization subdirectory of the CDGA Align or CDGA NoAlign output folder.

1. Pyranose atom coordinates XYZ files

A separate [molecule_name]_host_pyranose.XYZ file is generated in C CDGA_Tool_processed_molfiles\CDGA_Align_molfiles\Host_characterization\ directory for each processed structure. The XYZ file contains the atoms of the pyranose rings in a consistent order suitable for subsequent Cremer-Pople ring puckering analysis. For every pyranose unit, the atoms are written in the order:

O_{ring}, C1, C2, C3, C4, C5.

Consequently, the XYZ file contains 36 atoms for α -CD, 42 atoms for β -CD, and 48 atoms for γ -CD. Although the atom order in the generated output files is adjusted to ensure consistent pyranose ordering between complexes, the atom labels retain the original labels from the corresponding MOL file, allowing the reported atoms to be traced back to the input structure.

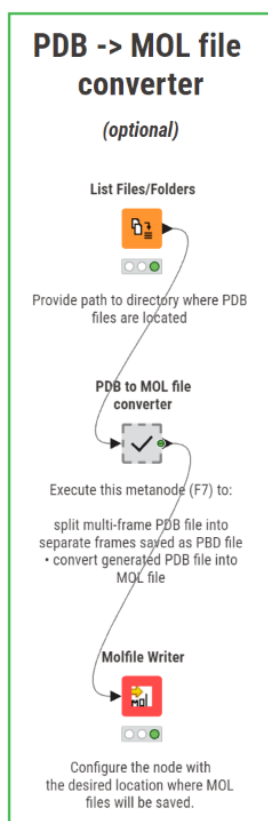
2. Glycosidic torsion angle file

The file **host_torsion_angles.csv** contains the glycosidic ϕ and ψ torsion angles for each pyranose linkage.

8.1. Host Torsion Angles – output table

Column name	Description	Meaning	Possible options or examples
entry_name	Original row identifier from the input table	Used to keep the link between the input structure and the calculated results	row ID from previous nodes, usually MOL file name
pyranose_i_atoms	Atom identities assigned to pyranose unit <i>i</i> and used for calculation of the glycosidic torsion angles. The field reports H1, C1, C4, O4, and H4 using the atom labels from the corresponding MOL file.	Identifies the atoms used to define the torsion angles associated with pyranose unit <i>i</i> and its glycosidic linkage. Label H1, C1, C4, O4, and H4 denote the standard atom positions in the pyranose unit.	H1=atom label corresponding to H1 in MOL file; C1=...; C4=...; O4=...; H4=... Example: H1=H10; C1=C1; C4=C7; O4=O16; H4=H8
phi(i)	Glycosidic ϕ torsion angle for the linkage from pyranose <i>i</i> to pyranose <i>i</i> +1.	Describes the conformation of the glycosidic linkage using the atoms H1(<i>i</i>), C1(<i>i</i>), O4(<i>i</i> +1), and C4(<i>i</i> +1).	Numeric value in ° example: phi1 = -73.4281°
psi(i)	Glycosidic ψ torsion angle for the linkage from pyranose <i>i</i> to pyranose <i>i</i> +1.	Describes the same glycosidic linkage using C1(<i>i</i>), O4(<i>i</i> +1), C4(<i>i</i> +1), and H4(<i>i</i> +1).	Numeric value in ° Example: psi1 = 112.7645°
phi(n)	Glycosidic ϕ torsion angle for the final linkage, which closes the cyclic CD structure by connecting pyranose <i>n</i> to pyranose 1.	Defined by H1(<i>n</i>), C1(<i>n</i>), O4(1), and C4(1).	Numeric value for β -CD, phi7; for α -CD, phi6; for γ -CD, phi8.
psi(n)	Glycosidic ψ torsion angle for the linkage from pyranose <i>n</i> to pyranose 1.	Describes the same glycosidic linkage using C1(<i>n</i>), O4(1), C4(1), and H4(1).	Numeric value for β -CD, psi7; for α -CD, psi6; for γ CD, psi8.

9. Appendix 2: PDB → MOL file converter



CDGA uses MOL files as input. To support PDB structures, including multi-frame PDB files, a separate conversion workflow is provided. The workflow uses an Open Babel node and therefore requires installation of the Vernalis KNIME Nodes extension.

To run the workflow:

1. Configure the first node **List Files/Folders** with the path to the directory containing PDB files. Multiple files can be processed together.
2. **Execute.**
3. Execute the next metanode to split PDB files into individual frames and convert them to MOL files. Temporary PDB files are stored in:

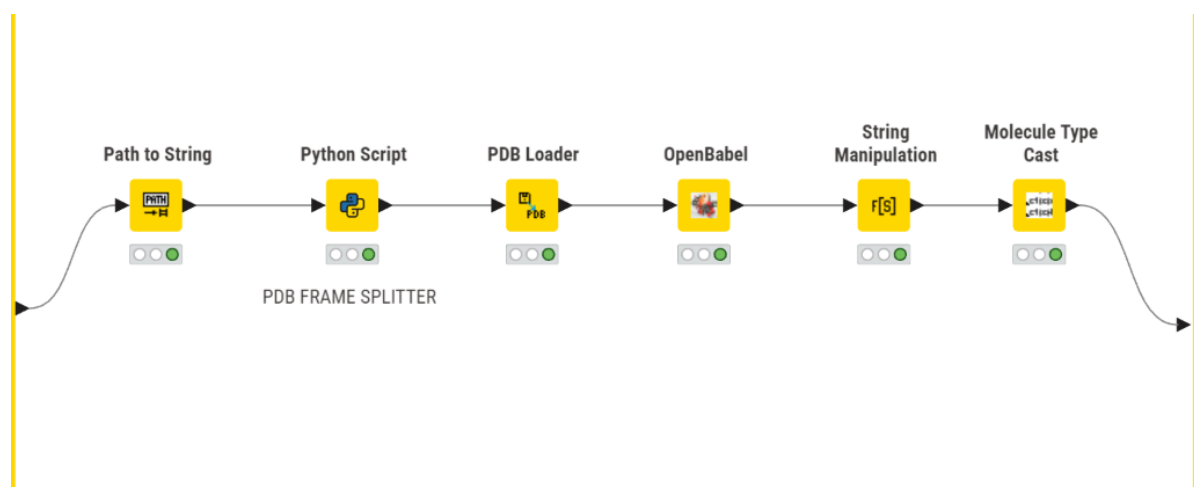
"Temp\CDGA_Tool_processed_molfiles\PDB_CONVERTER\"

The file name is based on the original file name, numbered using the 000n convention. The first line in the new PDB file is consistent with the name of the file. All created PDB single-frame files are grouped into a directory with the name of the original file.

4. Configure the last node with the desired output directory and execute it to save the MOL files.

The converter is kept separate from the main CDGA workflows so that the generated MOL files can be checked before analysis, since PDB files may contain incomplete bond information.

Metanode content:



10. Glossary

Term	Definition
CD centroid	Geometric centroid of all atoms belonging to the largest molecular fragment in the MOL file, assumed to be the CD host. Explicit hydrogen atoms are included and all atoms have equal weight. Used as the origin of the initial CD coordinate frame.
Z-axis	Axis defined by the vector from the narrow rim centroid to the wide rim centroid: $z = \text{normalize}(C_{\text{wide}} - C_{\text{narrow}}).$
X-axis	Principal direction obtained by PCA of the narrow rim atoms after their coordinates are projected onto the plane perpendicular to the z-axis.
Y-axis	Axis calculated as ($y = \text{normalize}(z \times x)$). Completes the orthogonal CD coordinate frame, perpendicular to z- and y-axes.
Initial CD coordinate frame	Orthogonal coordinate system defined by the CD centroid as origin and the X, Y, and Z axes described above. Used for the initial alignment of the tested CD complex to the reference CD. Two orientations differing by 180° around the cavity axis are evaluated and the one giving better rim correspondence is selected.
Narrow rim centroid	Geometric centroid of atoms identified as belonging to the primary rim, these are the C and O atoms matched by [CH2][OX2]. Hydrogen atoms are excluded. It represents the geometric centre of the narrow rim of the cavity.
Wide rim centroid	Geometric centroid of atoms identified as belonging to the secondary rim, these are the C and O atoms matched by [CH][OX2]. Hydrogen atoms are excluded. It represents the geometric centre of the wide rim of the cavity.
Cavity midpoint	Point exactly halfway between the narrow and wide rim centroids: $\left(M = \frac{C_{\text{narrow}} + C_{\text{wide}}}{2}\right).$
CD axis	Line connecting the narrow and wide rim centroids and passing through the cavity midpoint.
Kabsch refinement	Structural refinement performed after the initial coordinate frame alignment using matched heavy atoms of the cyclodextrin core. It provides the final host superposition after the approximate position and orientation have already been established from the CD coordinate frame.
Guest centroid	Geometric centroid of all heavy atoms of the guest. Hydrogen atoms are excluded and all heavy atoms have equal weight. It is used for guest depth calculations.

Term	Definition
Projection of guest centroid	Orthogonal projection of the guest centroid onto the CD axis.
Guest depth	Signed distance between the cavity midpoint and the projection of the guest centroid onto the CD axis. Positive values indicate displacement toward the narrow rim. Negative values indicate displacement toward the wide rim.
Local guest depth	Guest depth calculated using the cavity midpoint and CD axis of the current CD structure.
Reference guest depth	Guest depth calculated using the cavity midpoint and CD axis of the reference CD structure.
Guest axis	First principal molecular axis obtained by PCA of all heavy atom coordinates of the guest.
Guest angle	Acute angle between the PCA guest axis and the CD axis. Expected range is 0° to 90°. Values close to 0° indicate an approximately parallel orientation. Values close to 90° indicate an approximately perpendicular orientation.
ϕ torsion	Glycosidic torsion for linkage (<i>i</i>) to (<i>i</i> +1), defined by H1(<i>i</i>), C1(<i>i</i>), O4(<i>i</i> +1), C4(<i>i</i> +1). Pyranose units in every tested structure follow the sequence established for the reference host.
ψ torsion	Glycosidic torsion for linkage (<i>i</i>) to (<i>i</i> +1), defined by C1(<i>i</i>), O4(<i>i</i> +1), C4(<i>i</i> +1), H4(<i>i</i> +1). Pyranose units follow the same reference-based sequence. The final ψ value describes the linkage between the last and first pyranose units.
Cartesian X, Y, Z coordinates	Cartesian coordinates of the aligned pyranose ring atoms. Pyranose units in each tested structure are reported in the same sequence as in the reference host. Within each pyranose unit, the atom order is O_ring, C1, C2, C3, C4, C5. Coordinates can be used for Cremer-Pople puckering analysis.

11. Revision history

Version 1.0 – initial version, released on 15-Sep-2026, containing Host Characterization Module (Appendix 1) and PDB converter (Appendix 2) and Glossary.

Note:

This manual was reviewed with the assistance of AI to improve language and clarity. The authors reviewed and verified the complete text for accuracy. The picture on the front cover was generated with the assistance of AI (OpenAI, ChatGPT 5.5) and subsequently reviewed by the authors for its scientific accuracy.